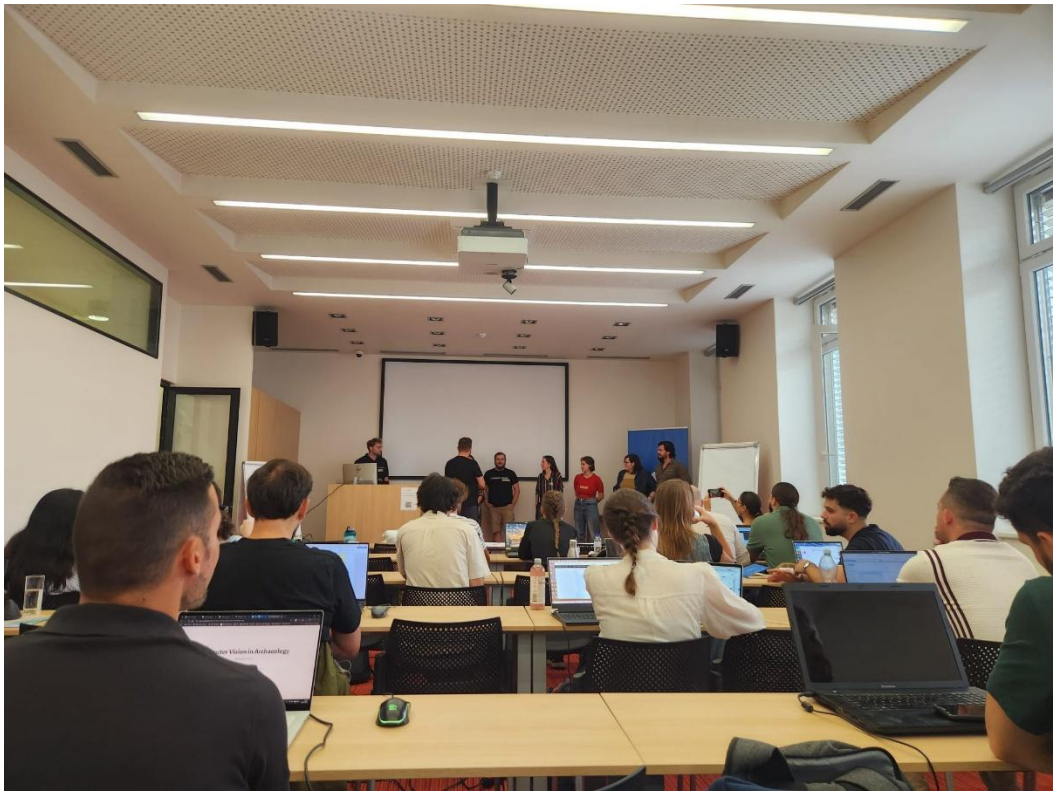
Computer Vision in Archaeology Training School in Brno, Czech Republic

In September 2026, I travelled to Brno, Czech Republic, to attend a five-day training school on Computer Vision in Archaeology, organized through the ATRIUM programme and hosted by the Institute of Archaeology of the Czech Academy of Sciences in Brno.

I came to the course with some previous experience in predictive modelling. During my PhD, I used linear models in R to estimate fish size from archaeological bone measurements. The training school opened my eyes to a much wider range of community resources now available for archaeological research.

It was also a great opportunity to meet archaeologists at different career stages who shared an interest in exploring how these emerging methods could contribute to their research. Knowing how many people had applied made me especially grateful to attend, and I was glad to hear that more ATRIUM training opportunities are being planned.

The course was particularly relevant to my work at the Material Culture PaleoLab at the University of Haifa, where we work with large lithic assemblages. Computer vision and predictive modelling have clear potential to support their classification and analysis, and I returned with exciting new ideas on how these approaches could become part of our future research.



The Training School

Day 1

Monday began by meeting the organizers and speakers, including Petr, Ronald, Filip, Anastacia and Ashley, and it was clear how much preparation had gone into the week. A series of practical notebooks formed the backbone of the course, allowing us to follow the workflows ourselves, modify the code, occasionally break it, and return to them afterwards as a resource.

The first day introduced us to the basics of computer vision and the kinds of questions it can answer. An important principle was that while an archaeologist looks at an image and immediately brings in context such as period, style or cultural group, a computer initially sees pixels, brightness, edges and textures. We were also introduced to different computer-vision tasks, including classification, object detection, segmentation and keypoint detection, as well as discriminative and generative modelling approaches.

In the afternoon, we moved into AI-assisted coding, which was quite an adventure for someone used to working in R before the age of AI. I chose to work in Google Colab, and we also explored tools such as Git, VS Code/Zed and coding agents.

The day ended with a fun icebreaker with lots of Czech beer, vegan snacks, and a chance to get to know the other participants.



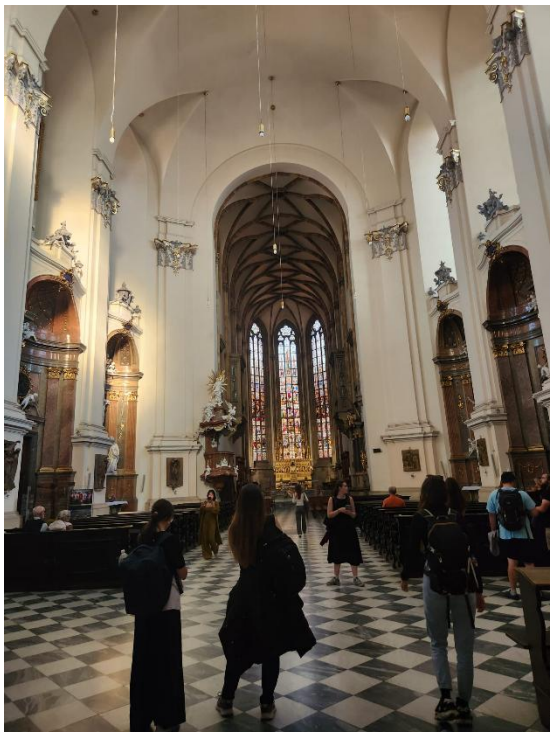
Photo by David and Zuzana

Day 2

Tuesday focused on preparing archaeological images for machine learning. We learned how 2D images can be annotated and organized into structured training data, using tools such as CVAT and formats such as COCO. We also discussed the challenge of keeping archaeological categories consistent across datasets when labels, researchers, photography and classification can all vary

In the afternoon, participants presented their own projects and datasets ranging from remote sensing and image analysis for grave detection to coin imaging and biomechanical studies of stone-tool production. It was reassuring to see that very different projects shared similar challenges, including limited sample sizes, variable image quality and difficult classifications.

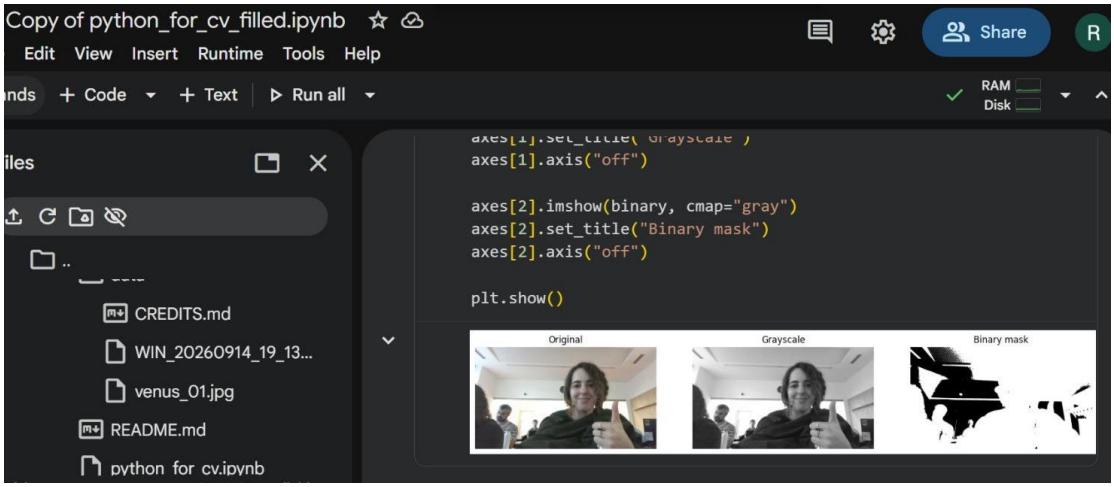
Tuesday evening took us out of the classroom for a walking tour of Brno. We visited the Old Town Hall and its famous "Brno dragon," as well as several of the city's historic churches. I especially enjoyed the Cathedral of St. Peter and Paul, with its Baroque interior and Gothic and Neo-Gothic exterior.



Day 3

Wednesday focused on microscopy, image preprocessing, standardization and data augmentation. We learned how factors such as lighting, contrast, resolution and colour can make archaeological surface look very different from one image to another, and why images often need to be standardized before they can be compared reliably.

We then worked through practical examples using techniques such as resizing, normalization, histogram matching, edge and texture detection, and data augmentation. The microscopy case studies included bone surface modifications, micromorphology and lithic use-wear, the last of which was especially relevant to my own lab work. We also had some fun experimenting with our own images in the course notebooks.



Day 4

Thursday's topic was satellite imagery and archaeological remote sensing.

We were introduced to multispectral imagery and how different surfaces reflect and absorb different wavelengths of light. We then used these data to look for archaeological cropmarks and trained a Random Forest classifier to identify possible features. It was useful to see how features such as field boundaries and agricultural tracks could initially be misidentified as archaeological targets, but these false positives could then be added back into the training data to help the model distinguish them more effectively. The final workflow combined satellite data, image processing, machine learning and GIS interpretation, showing how all of these steps work together to identify and classify possible archaeological features.

In the afternoon, we switched to rock art. We worked with enhanced visualizations generated from 3D models and learned how large images can be divided into smaller tiles so that subtle features are not lost when the image is resized.

In the evening, a rainstorm effectively kept us inside the institute's recreation room after another icebreaker and even some bards and singing from our talented participants. we were not so worried and became increasingly confident dart players, and there was enough good Czech beer for everyone.

Day 5

The final day focused on the coins dataset and brought together many of the ideas from the week. The challenge was how to compare large numbers of archaeological coins, where corrosion, damage and surface variation can make straightforward image comparison unreliable. We were shown an approach adapted from astronomy, where distinctive points in the coin designs were treated like stars in a constellation and compared according to their spatial arrangement. I loved this example because it was a great demonstration of creative problem solving, and also quite whimsical. It felt like a fitting topic to end the week on.

Closing Thoughts

The training school in Brno gave me a much clearer understanding of what computer vision and machine learning can contribute to archaeological research. It also reminded me how productive crossing disciplinary boundaries can be. An archaeological problem may have an unexpected parallel in astronomy, remote sensing, computer graphics or another field entirely. It was also great to meet so many likeminded researchers that I hope to stay in touch and collaborate with in the future.

Below, enjoying the Lithic tool display at the Moravian Regional Museum after the course

